AMD HACKATHON · 2026

BrainOS

The layer between scattered company knowledge and AI agents.

Multi-agent · Multi-modal · Knowledge graph · AMD MI300X

DIRECTED GRAPH | SUPERSESSION | GAP DETECTION | MULTIMODAL VLM | SLACK-NATIVE | SELF-HOSTABLE

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github.com/vamshinr/BrainOS • Live: 129.212.176.117:3000

THE PROBLEM

Every company has a hidden second corpus.

> *"I think every company in the world is going to need one." — Tom Blomfield, Y Combinator · RFS: Company Brain*

Where knowledge actually lives

- Slack threads that scroll into history
- PDFs updated by email, not the doc
- Whiteboard photos in personal drives
- Architecture diagrams 6 months stale
- Three engineers who remember the decision
- Contracts that contradict the runbook

What happens to AI agents today

✗ No context → takes the wrong action
✗ Stale policy → legal / compliance liability
✗ Missing owner → ticket stuck for days
✗ Conflicting docs → LLM hallucinates a merge
✗ Can't read diagrams → blind to system design
✗ No output format → agent can't load it

The models crossed the quality bar in 2024. The knowledge layer is what's still missing.

THE SOLUTION

BrainOS is the missing knowledge layer.

Ingest anything

Text, PDF, DOCX, images, architecture diagrams, Slack threads, slash commands. One unified pipeline regardless of format.

Reconcile, don't stack

Every new fact is compared against the existing graph. Old facts are **superseded**. Contradictions flagged **Disputed** — not silently buried.

Agent-ready output

Exports a SKILLS.md per department. Drop into Claude Code, Cursor, OpenAI GPTs, or Aider — the agent operates with full company context.

PIPELINE

Slack/PDF/DOCX/Image → Ingestion Agent → Structuring Agent → Knowledge Graph → SKILLS.md → AI Agent

Every step runs on AMD MI300X via vLLM — Qwen 32B + Qwen2-VL 7B co-resident on a single card.

WHY THIS ACTUALLY MATTERS

Five things every company gets, on day one.

① **Conflict surfacing prevents the "we decided what?" problem.**

When two docs disagree, both get kept with a Disputed badge. The argument surfaces instead of being silently averaged by an LLM.

② **Knowledge gap analysis becomes a hiring / focus tool.**

missing_owner, undescribed_entity, open_dispute — a deterministic punch list of what the org is blind to. Drives priorities, not vibes.

③ **You don't need to write docs you'd never write anyway.**

Slack threads, whiteboard photos, PDFs already exist. BrainOS reconciles them into structured facts — no parallel doc culture required.

④ **Your AI agents become useful on day one.**

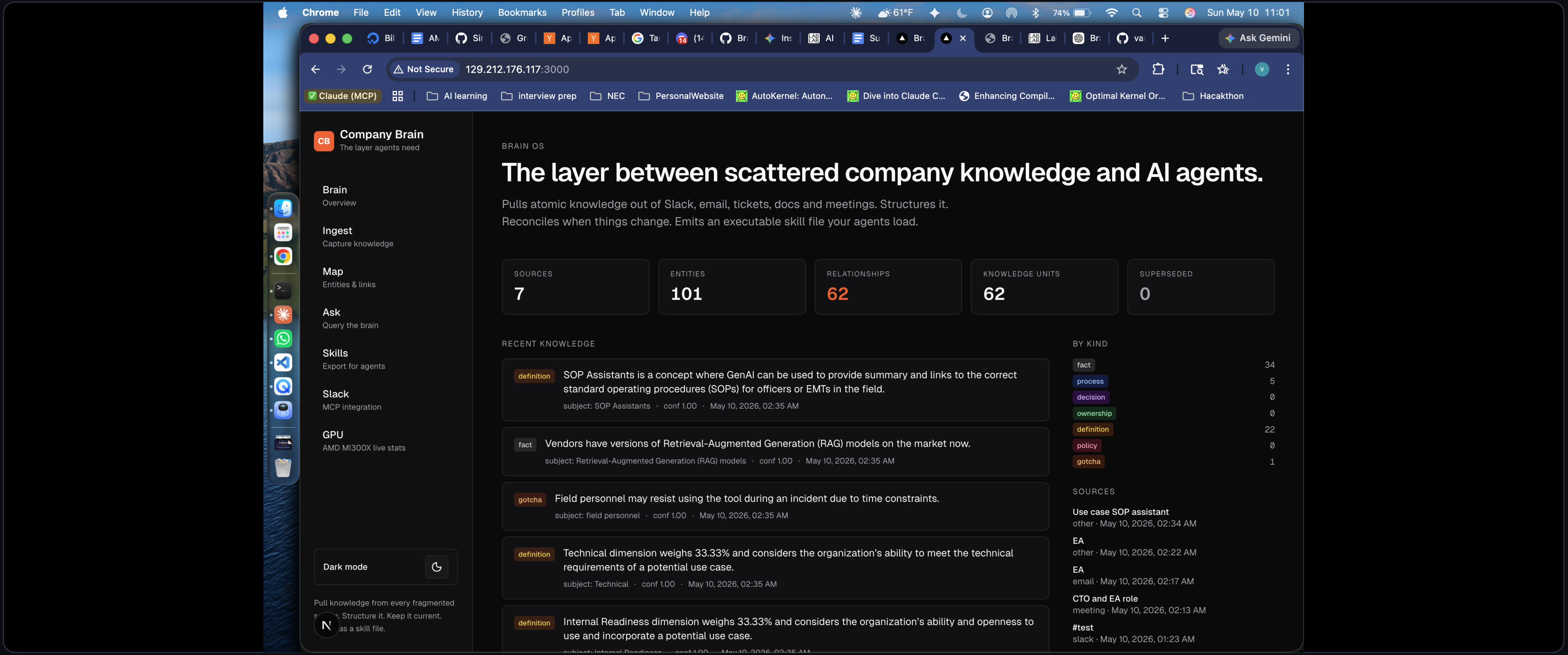
SKILLS.md drops into Claude Code, Cursor, GPTs, Aider. Agents inherit full company context — ownership, policies, gotchas, conflicts.

⑤ **You don't lose context when someone leaves.**

Maya's departure becomes an event the graph processes, not a fire drill. Ownership edges rewire; orphaned tasks surface as gaps.

WHAT IT LOOKS LIKE

Not a slide deck. A live product.



Dashboard at 129.212.176.117:3000 · 101 entities · 62 relationships · 62 knowledge units · 0 superseded after reconcile

DIFFERENTIATION

This is not RAG.

Vanilla RAG

- ✗ Embed → cosine → top-k chunks → LLM
- ✗ One retrieval signal (vector only)
- ✗ Keeps old + new policy **simultaneously**
- ✗ No graph — can't traverse ownership
- ✗ Blind to images, diagrams, whiteboards
- ✗ Hallucinates when sources conflict
- ✗ Says "yes" when it has no data

BrainOS

- ✓ 5-signal hybrid retrieval + RRF rerank
- ✓ Vector + BM25 + entity + graph + multimodal
- ✓ Temporal scoring — **one fact wins** per query
- ✓ Graph walk: Customer → CSM → escalation chain
- ✓ VLM reads diagrams, whiteboards, org charts
- ✓ Disputed badge — both kept, conflict exposed
- ✓ Gap analysis — **knows what it doesn't know**

ARCHITECTURE

Four specialized agents — not one mega-prompt.

① Ingestion

- Sentence-aware chunking
- Qwen2-VL 7B for images
- Atomic unit extraction
- Retry with backoff

② Structuring

- Embed → ChromaDB HNSW
- Reconcile: 4 verdicts
- validTo + supersededAt
- Merges into brain.json

③ Execution

- 5-signal hybrid retrieval
- Temporal rerank
- Stale filter
- Inline-cited answer gen

④ Feedback

- Groundedness audit
- Conf < 0.72 → rewrite
- Pre/post-revision logged
- Verdicts → /api/metrics

Per-task model routing: extraction → Qwen 32B · reconciliation → Qwen 32B · vision → Qwen2-VL 7B · feedback → Qwen 32B

RECONCILIATION IS THE CORE PRIMITIVE

Old facts get superseded, not stacked.

New unit extracted → find existing units with overlapping entities → Structuring Agent emits a verdict.

supersedes

Old unit → validTo=now, temporalStatus=historical. Disappears from "current" answers.

duplicate

New unit dropped. Source is cross-referenced on the existing unit.

conflicts

Both kept. conflictsWith back-reference. **Disputed** badge in UI.

independent

Stored as-is. No effect on existing units.

Time-travel queries

"What was the refund limit in April?" → temporal scorer boosts expired units.
"What is the refund limit today?" → temporal scorer penalizes historical units.
Both answers correct. Neither hallucinates.

Knowledge gap detection

Pure graph topology scan, no retrieval:

- **missing_owner** — entity with no ownership edge
- **undescribed_entity** — named but never defined
- **orphan_gotcha** — warning with no owning entity
- **open_dispute** — unresolved conflictsWith pair

FEATURES IN ACTION

Four capabilities. One real query each.

All from the Helix Outdoor demo brain — same graph, four different retrieval modes.

Answer from graph

QUERY

Q: Who owns the Saigon TexCo factory relationship right now?

BRAINOS ANSWER

A: Marco Ferraro (acting). The fact lives in an *edge* (Marco → Saigon TexCo), not a doc. No vector embedding would ever retrieve this.

Temporal + supersession

QUERY

Q: CX Lead refund limit today? · And in April 2026?

BRAINOS ANSWER

A today: **$1,000** (eff 2026-05-01). A April: **$2,000**. Old policy marked [SUPERSEDED], not deleted — both answers correct, neither hallucinated.

Multimodal (VLM)

QUERY

Q: Looking at the architecture diagram, what's the most fragile webhook?

BRAINOS ANSWER

A: The **ShipBob webhook** — VLM reads the diagram, fuses with a Slack thread about a missing HMAC signature check on retries.

Knowledge gap analysis

QUERY

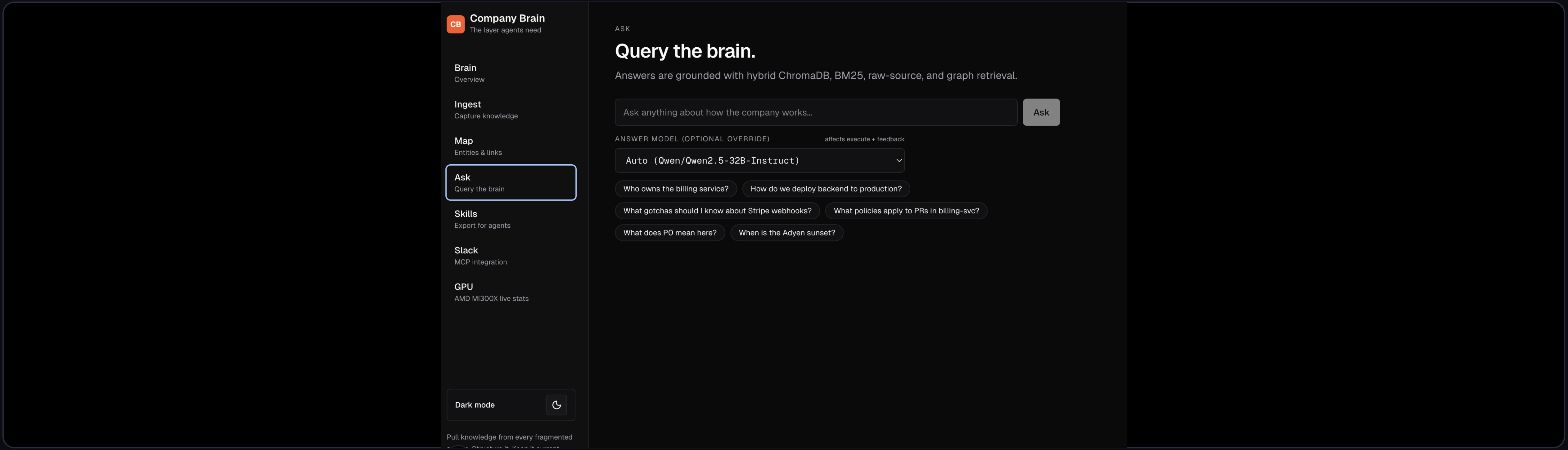
Q: What doesn't Helix Outdoor know about itself?

BRAINOS ANSWER

A: 5 gaps — 2 missing_owner, 1 undescribed_entity, 1 orphan_gotcha, 1 open_dispute. Deterministic graph scan — no LLM call.

LIVE · /ASK

Grounded answers · retrieval diagnostics · per-answer audit.



Grounded

Every answer carries a grounding badge, confidence (0-1), and inline source citations [F1], [R2].

Diagnostic

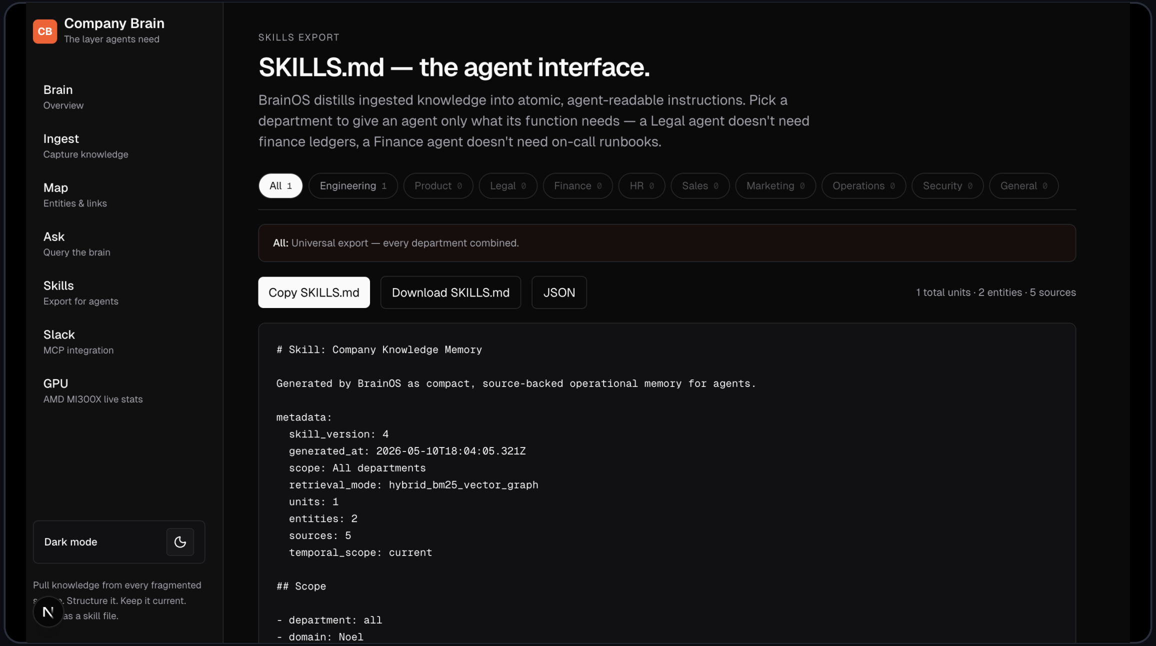
Retrieval debug pane shows all 6 signal types — vector, BM25, entity, graph — per query.

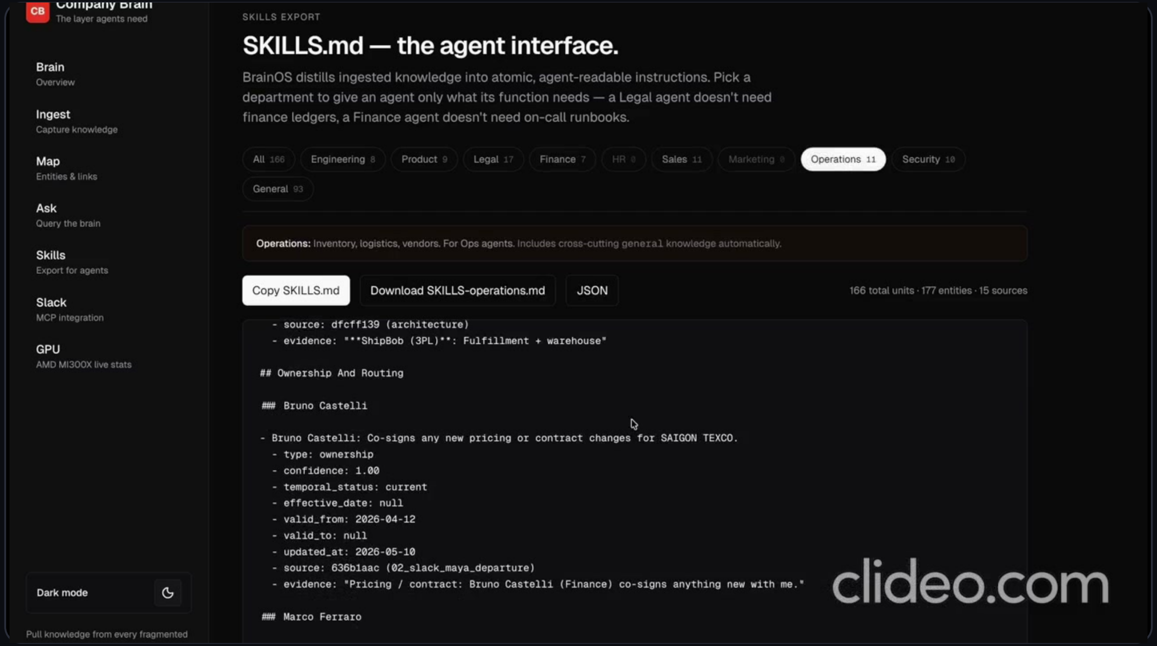
Honest

"Brain does not have this" instead of hallucinating. Draft vs revised answer disclosed when verifier rewrites.

OUTPUT FORMAT

SKILLS.md — the portable company brain.





What's in a SKILLS.md

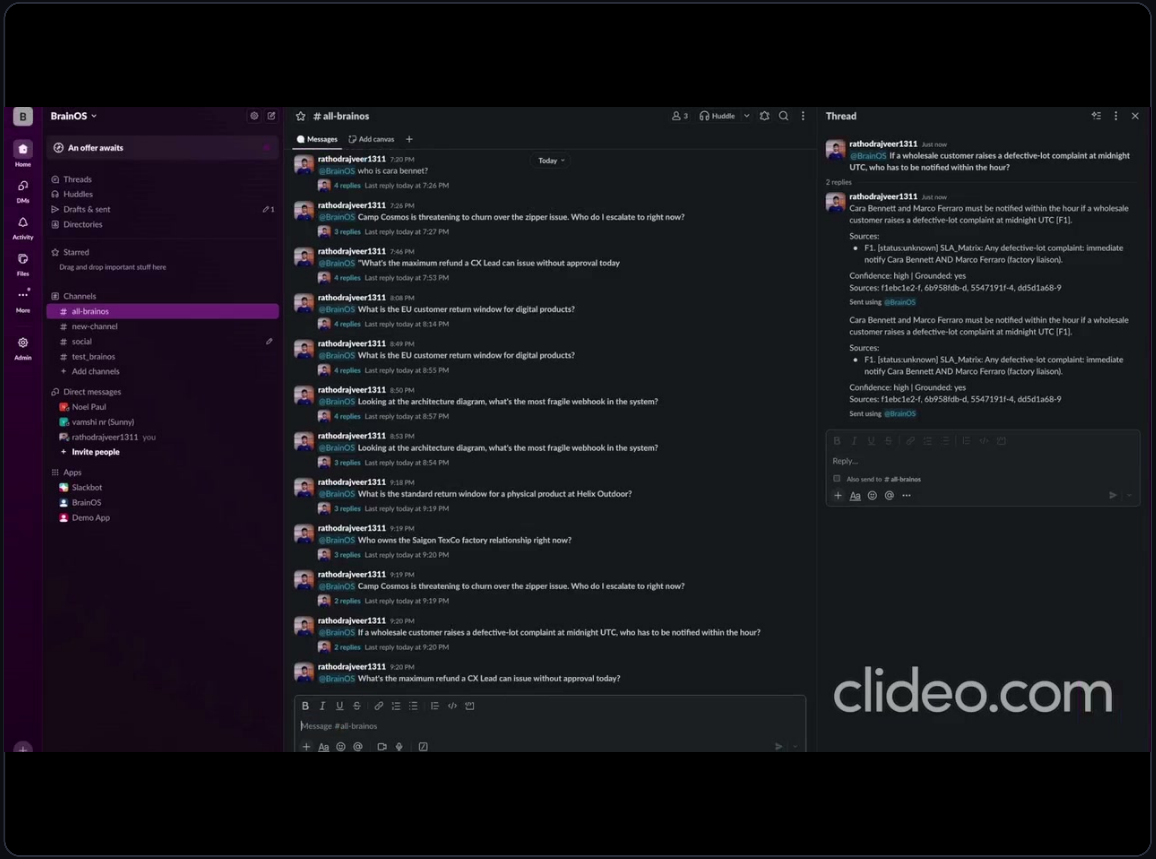
Scope · Operational Facts · Ownership & Routing · Active Policies · Processes · Gotchas · Decisions · Temporal Notes · **Agent Rules** (≥0.75 conf) · Graph Relationships · Source Index. Scoped per department — Legal doesn't see Finance.

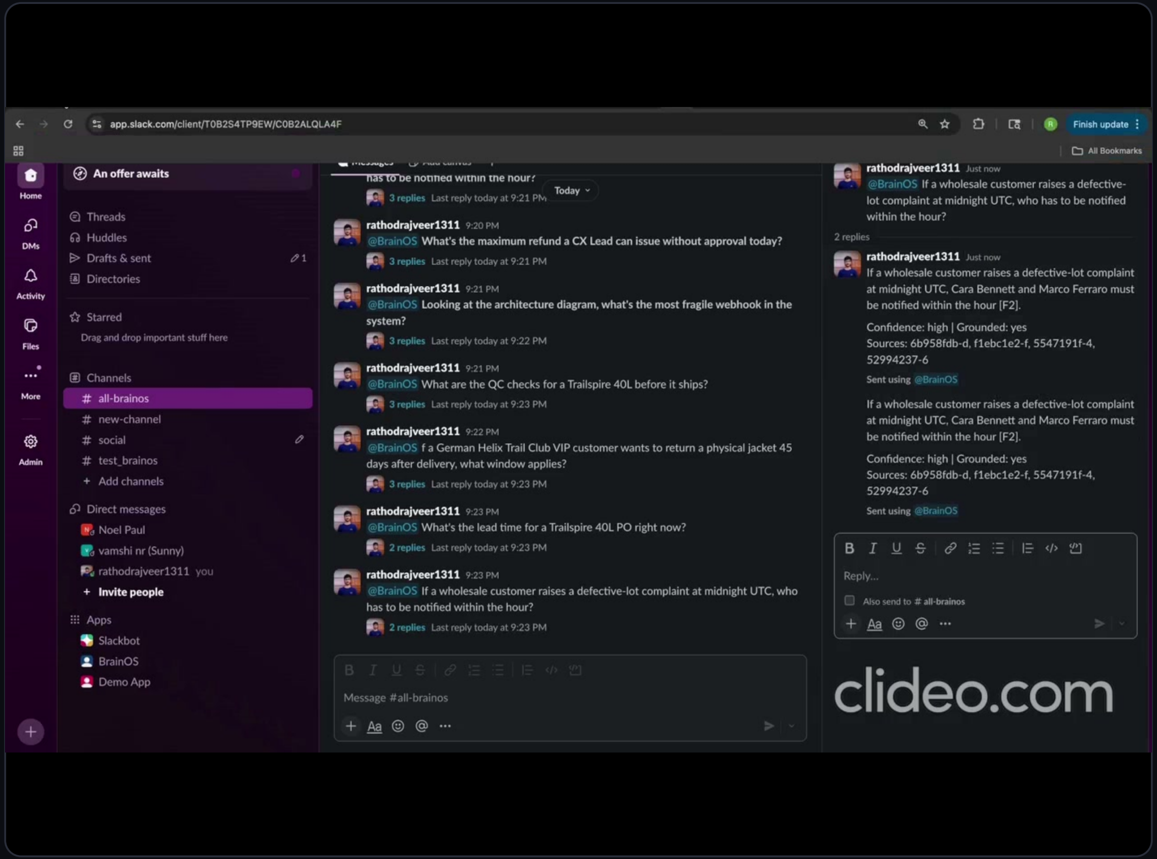
Sample Agent Rules

```
- Webhook: missing X-Shipbob-Signature MUST 400.
- Lot 24-A-118: DO NOT restock; full refund.
- Saigon TexCo → Marco Ferraro (acting).
- CX Lead refund cap = $1,000 (eff 2026-05-01).
```

SLACK · IN ACTION

Grounded answers, in-thread, with sources.



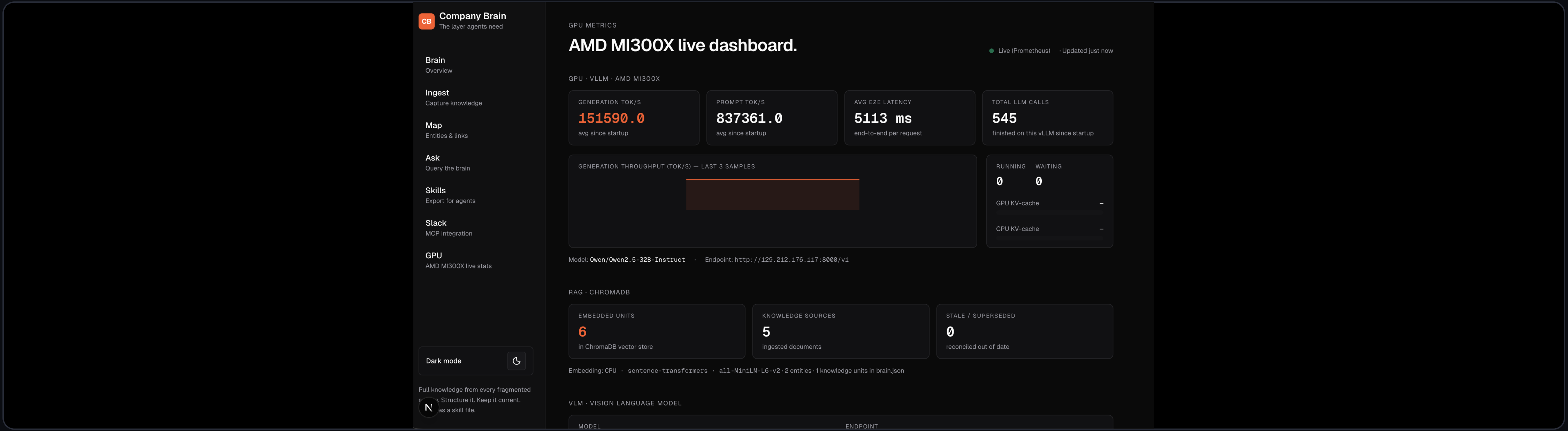


Real questions in #all-brainos: defective-lot SLA, Saigon TexCo ownership, refund limits, architecture webhooks.
Every reply cites F1, F2, source IDs, confidence label. HMAC signature verification on every request.

WHY AMD MI300X IS THE RIGHT HARDWARE

192 GB HBM3 — co-resident text + vision on one card.

192GB	1 GPU	<2s
HBM3 — CO-RESIDE QWEN 32B + QWEN2-VL 7B	VS 3× H100 FOR THE SAME WORKLOAD	END-TO-END ANSWER LATENCY ON VLLM



Live /metrics · Prometheus scrape · gen tok/s · prompt tok/s · KV-cache util · per-call log of every LLM call.

COMPETITIVE LANDSCAPE

The capabilities no incumbent ships.

CAPABILITY	GLEAN	NOTION AI	COPILOT	BRAINOS
Reconciliation / supersession	✗	✗	✗	✓
Directed knowledge graph	✗	✗	✗	✓
Conflict surfacing (Disputed)	✗	✗	✗	✓
Knowledge gap detection	✗	✗	✗	✓
VLM multimodal ingestion	✗	✗	PARTIAL	✓
Agent-loadable SKILLS.md	✗	✗	✗	✓
Self-hostable / open source	✗	✗	✗	✓

Who needs this first: regulated industries (defense, healthcare, EU fintech) · AI-native startups · ops-heavy scale-ups.

Five minutes from a cold codebase to an **agent that knows the company.**

That's a company brain. Built on AMD MI300X. Open source. Ready for the next 99 companies.

LIVE DEMO

`129.212.176.117:3000`

SOURCE

`github.com/vamshinr/BrainOS`

AMD MI300X · OPEN SOURCE · SELF-HOSTABLE · VENDOR-NEUTRAL · TRACK 1 + 3

Noel · Rajveer · Vamshi · AMD Hackathon 2026